NLP Mid Exam 2025/2026

1. **Which of the following best distinguishes lemmatization from stemming in NLP?**
   A. Lemmatization uses predefined suffix rules, while stemming depends on vocabulary lookup.
   B. Stemming uses context-based transformations, while lemmatization simply truncates words.
   C. Lemmatization preserves the grammatical and morphological analysis, whereas stemming does not.
   D. Stemming always produces valid dictionary words, while lemmatization may not.
2. **Why is the Continuous Bag-of-Words (CBOW) model considered faster but less effective for rare words compared to Skip-gram?**
   A. It ignores the target word and predicts its surrounding context words.
   B. CBOW averages multiple context words to predict one target, reducing computational complexity but losing rare word distinctions.
   C. CBOW uses a higher-dimensional hidden layer that overfits frequent words.
   D. CBOW requires more training for rare words, increasing noise.
3. **In Word2Vec training, what happens when the window size parameter is increased?**
   A. It reduces the number of training samples generated.
   B. It captures a broader range of contextual words, thus capturing broader semantic relationships.
   C. It increases overfitting by limiting context.
   D. It prevents the model from learning syntactic relations.
4. **What is a fundamental computational drawback of Bag-of-Words (BoW) and One-Hot Encoding that Word Embeddings overcome?**
   A. They cannot handle non-English vocabularies.
   B. They produce high-dimensional sparse vectors that ignore semantic relationships between words.
   C. They are labeled data for training word representations and rely on deep neural networks for vectorization.
   D. They introduce biases from word frequency.
5. **In a Recurrent Neural Network (RNN), what is the primary role of the feedback loop?**
   A. To propagate data forward only.
   B. To enable the network to remember previous inputs.
   C. To increase the number of output neurons.
   D. To reduce the number of input connections.
6. **When unrolling an RNN through time, what is a key effect on the network structure?**
   A. It converts the RNN into a feedforward network with multiple inputs and outputs.
   B. It removes the biases from the network.
   C. It consolidates multiple layers into one layer.
   D. It eliminates the need for activation functions.
7. **Which of the following best describes the purpose of Word Embeddings in NLP?**
   A. To represent text using word frequency counts only.
   B. **To convert words into numeric vectors that capture semantic and syntactic meaning**.
   C. To create sparse matrices for text classification.
   D. To store each word as a unique binary index.

8. **What is a major limitation of the Bag-of-Words (BoW) model?**
   A. It captures the semantic relationships between words.
   B. It requires labeled data for training.
   C. It ignores the order and context of words in a document.
   D. It reduces dimensionality automatically.
9. **Which of the following best describes the difference between RNNs and standard feedforward neural networks?**
   A. RNNs include feedback loops that enable temporal dynamic behavior.
   B. RNNs do not use activation functions.
   C. Feedforward networks are unrolled across time unlike RNNs.
   D. RNNs do not have weights and biases.
10. **In the Skip-gram model, what serves as the input and what is predicted as output?**
   A. Input: context words, Output: target word.
   B. Input: document, Output: topic.
   C. Input: target word, Output: context words.
   D. Input: frequency count, Output: embedding vector.

NLP Quiz 2025/2026

1. **What is the main idea behind Word2Vec?**
   A. Convert words into one-hot vectors
   B. **Convert words into dense vectors based on context**
   C. Convert sentences into images
   D. Count the frequency of words
2. **Which of the following is NOT a training architecture of Word2Vec?**
   A. Skip-Gram
   B. CBOW
   C. **GloVe**
   D. Both A and B
3. **What is a unique advantage of FastText over Word2Vec?**
   A. It trains faster
   B. **It represents words as bags of characters (sub-word units)**
   C. It produces larger embeddings
   D. It only works on English
4. **Why does FastText handle rare words better than Word2Vec?**
   A. It ignores rare words
   B. **It uses subword information**
   C. It increases embedding size
   D. It uses more epochs
5. **What problem do RNNs commonly suffer from?**
   A. Overfitting
   B. **Vanishing and exploding gradients**
   C. Slow inference
   D. Memory overflow

6. **Which mechanism allows LSTMs to overcome the vanishing gradient problem?**
   A. Large batch size
   B. One-hot encoding
   C. **Gated memory cells (forget, input, output gates)**
   D. More layers
7. **What is the main advantage of LSTM over a simple RNN?**
   A. It trains faster
   B. **It stores long-term dependencies better**
   C. It reduces dataset size
   D. It never overfits
8. **Which of the following is true about RNNs?**
   A. They process all inputs independently
   B. **They maintain a hidden state that carries information across time**
   C. They can only process images
   D. They don't use activation functions
9. **In Word2Vec Skip-Gram, the model predicts:**
   A. The word from its characters
   B. **The context words given the target word**
   C. The next sentence
   D. The next character
10. **FastText is particularly useful for:**
   A. **Languages with complex morphology**
   B. Short English sentences only
   C. Tasks requiring image embeddings
   D. Noise reduction in audio